

Alexander Urs-Badet

Full-Stack Software Engineer | Computer Vision & Applied AI Systems

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SUMMARY

Full-stack software engineer with production experience building Angular/AWS systems for federally regulated environments serving 1,500+ users. Hands-on applied AI projects across real-time computer vision, pose estimation, robotics tracking/control, algorithmic backtesting, and server-authoritative game systems. Focused on well-tested systems that hold up in production, end to end.

TECHNICAL SKILLS

AI/ML Frameworks	PyTorch, TensorFlow, scikit-learn, OpenCV, DeepLabCut, Bonsai
Generative AI	Claude API, OpenAI, prompt engineering
Computer Vision	Real-time object detection, pose estimation, behavior tracking, action inference
Backend & Cloud	Python, Java, C#, Node.js, AWS (Lambda, S3, API Gateway), REST APIs
Frontend & Tools	Angular, React, JavaScript, Jest, Git, Agile/Scrum
Specializations	Real-time computer vision, pose estimation, algorithmic backtesting, system migration

PROFESSIONAL EXPERIENCE

Software Engineer | TSPi | Lynnwood, WA · 11/2023 – Present

- Led full-stack development of production systems serving 1,500+ FSA loan officers and applicants across the U.S. Farm Service Agency.
- Architected and deployed a financial optimization calculator (AWS Lambda, Node.js, Angular) that reduced manual calculation overhead by ~40%.
- Modernized legacy Java/Struts workflows into Angular/Spring Boot features while maintaining extensive Jest coverage across critical front-end components.
- Developed features for a nationwide debt-consolidation system presented to senior FSA and congressional stakeholders.

Software Engineering Intern | T-Mobile | Bellevue, WA · 07/2021 – 08/2021, 07/2022 – 08/2022

- Optimized core sorting and data processing algorithms used in Sync-Up product; achieved 50% runtime reduction.
- Built Python/Flask applications integrating external APIs; consolidated multiple fragmented data trackers into unified application.
- Mentored 2 junior interns in Python; enabled both to successfully deliver 2 projects independently, raising internal engineering capacity.

PROJECTS

Pose Estimation with DeepLabCut · Python, DeepLabCut, PyTorch, ResNet-50

- Labeled a 6-keypoint skeleton on ~250 k-means-sampled frames and trained a ResNet-50 pose network in DeepLabCut (PyTorch backend).
- Ran inference (analyze_videos) to produce per-frame keypoints and likelihoods, reduced to a confidence-weighted body centroid for downstream tracking.

Real-Time Tracking & Robot Control · Python, OpenCV, Kalman Filters, Robotics

- Built an OpenCV detection pipeline (LAB/HSV color segmentation, morphology, contour centroids) on a threaded 30 FPS capture loop that never tracks a stale frame.
- Wrote per-object constant-velocity Kalman filters with gated association and occlusion handling, then closed the loop onto a mobile robot over a hardware interface.

Algorithmic Trading System · Python, Options, Backtesting

- Built an automated intraday options trading system: pluggable signal engines (VWAP-pullback, opening-range breakout, mean-reversion) over intraday bars and options chains.
- Engineered a multi-layer risk engine (ATR/premium position sizing, daily-loss lockouts, cooldowns, throttles, kill switch) and broker-agnostic execution (Webull/IBKR/Tradier/paper), validated with a walk-forward backtester.

EDUCATION

- A.A. & Full Stack Developer Certificate · Edmonds College, Edmonds, WA · 2022–2023
- B.S. Psychology Coursework · University of Washington, Seattle